CHAPTER 3 & CHAPTER 4

RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

MATHEMATICAL EXPECTATION

RANDOM VARIABLES (RV)

DISCRETE RANDOM VARIABLES (PMF / CDF / EXPECTATION / VARIANCE)

CONTINUOUS RANDOM VARIABLES (PDF / CDF / EXPECTATION / VARIANCE)

JOINT PROBABILITY DISTRIBUTIONS

COVARIANCE & CORRELATION

RANDOM VARIABLES

RANDOM VARIABLES: INTRODUCTION

- Suppose you are a quality manager at a chips manufacturing plant. The quality of a batch is checked by picking 3 bags at random and checking the quality of each one. The sample space for this experiment can be expressed as:

$$S = \{PPP, PPF, PFP, FPP, FFP, FPF, PFF, FFF\}$$

- As the quality manager, one natural concern would be: "how many bags fail the quality test?"
- To each element in the sample space, we can assign a number that represents the number of failed bags:

S	PPP	PPF	PFP	FPP	FFP	FPF	PFF	FFF
X								

RANDOM VARIABLES

- A **random variable (RV)** is a real – valued function defined over a sample space. A RV assigns a value to each outcome in the sample space.
  - Typically, capital letters at the END of the alphabet are used to denote a RV.
  - The corresponding lower case letter is then used to denote a specific value.

RANDOM VARIABLES (CONTINUED)

- $\boldsymbol{P(X = x)}$ – The probability that X takes on the value x is defined as the sum of the probabilities of all sample points in S that are assigned the value x. We will sometimes denote $P(X = x)$ by $f(x)$.

- Recall the previous example where $X =$ "number of failed bags." Find the probability of each event occurring.

$\boldsymbol{S}$	$\boldsymbol{PPP}$	$\boldsymbol{PPF}$	$\boldsymbol{PFP}$	$\boldsymbol{FPP}$	$\boldsymbol{FFP}$	$\boldsymbol{FPF}$	$\boldsymbol{PFF}$	$\boldsymbol{FFF}$
X								
P								

- Find $P(X = x)$ for each outcome for X.

- Notice these probabilities sum to 1. Why?

- <u>Note</u>: In this example, S is finite which might not always be the case.

DISCRETE RANDOM VARIABLES

DISCRETE RANDOM VARIABLES

- A **discrete random variable** can assume only a finite or countably infinite number of values (counting numbers).
- Examples:
  - X = number of defectives
  - Y = number of hairs on your head
  - W = number of injury claims / month for an insurance company

- Most often interest lies in how the probabilities are distributed over all possible values of the sample space. This probability distribution can be given in the form of a:
  - Table, Graph, or Formula

EXAMPLE 1

- Workers consist of 3 men and 3 women. Suppose 2 workers are chosen at random. Note wanting to show bias, the foreman is interested in $X =$ number of women selected. Find the probability distribution of X.
  - What are the possible values of X?
  - How many elements are in S?
  - Find the probability of each outcome in X occurring.

EXAMPLE 1 (CONTINUED)

- On the previous slide, we found: $P(X = 0) = \frac{1}{5}$, $P(X = 1) = \frac{3}{5}$, and $P(X = 2) = \frac{1}{5}$.
- We can summarize the probabilities in the following ways. (note that each of these methods lists all the possible values for X and their corresponding probabilities):
- Table

EXAMPLE 1 (CONTINUED)

- On the previous slide, we found: $P(X = 0) = \frac{1}{5}$, $P(X = 1) = \frac{3}{5}$, and $P(X = 2) = \frac{1}{5}$.
- We can summarize the probabilities in the following ways. (note that each of these methods lists all the possible values for X and their corresponding probabilities):
- Graph

EXAMPLE 1 (CONTINUED)

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- We can summarize the probabilities in the following ways. (note that each of these methods lists all the possible values for X and their corresponding probabilities):
- Formula

PROBABILITY MASS FUNCTION (PMF)

- A **probability mass function (pmf)** is a set of ordered pairs $(x, f(x))$ of the discrete random variable X if, for all possible values of x:
  - $P(X = x) = f(x)$ ✓
  - $0 \leq f(x) \leq 1$ ✓
  - $\sum_x f(x) = 1$ ✓

EXAMPLE 2

- Find the pmf (in table form) for the following discrete random variable.
- A small child is asked to match each of three pictures of animals to the word identifying the animal. Suppose a child assigns the words to the pictures at random and let $X =$ number of correct matches.

EXAMPLE 3

- For each of the following, determine the constant c so that $f(x)$ satisfies the condition of being a pmf for a random variable X.
  - $f(x) = \frac{x}{c}$ where $x = 1,2,3,4$

- $f(x) = cx$ where $x = 1,2,3,,\ldots,10$

- $f(x) = c(x+1)^2$ where $x = 0,1,2,3$

CUMULATIVE DISTRIBUTION FUNCTION (CDF)

- The **cumulative distribution function (cdf)**, $\boldsymbol{F(x)}$, of a discrete random variable with pmf $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t) \text{ for } -\infty \leq x \leq \infty$$

EXAMPLE 4

- Find the cdf for the scenario given in Example 2.
  - A small child is asked to match each of three pictures of animals to the word identifying the animal. Suppose a child assigns the words to the pictures at random and let $X =$ number of correct matches.

- Find the cdf for the scenario given in Example 3a: $f(x) = \frac{x}{10}$ where $x = 1,2,3,4$

EXPECTED VALUE OF DISCRETE RV'S (AVERAGE)

- The **expectation** of a random variable X is a weighted average, denoted $\boldsymbol{E(X)}$, of the possible values that X can assume.

- Formal Definition: If X is a discrete random variable having probability mass function $f(x)$, then the expectation **(expected value or mean)** of X, denoted $E(X)$, is defined by:

$$\mu_X = E[x] = \sum_{f(x)>0} x \cdot f(x)$$

EXAMPLE 5

- Find the expected value of the pmf given in Example 3a: $f(x) = \frac{x}{10}$ where $x = 1,2,3,4$

- Find the expected value of the pmf given in Example 3c: $f(x) = c(x+1)^2$ where $x = 0,1,2,3$

EXAMPLE 6

- Let X denote a RV that takes on the values $-1, 0$, and 1 with respective probabilities

$$P(X = -1) = 0.2 \qquad P(X = 0) = 0.5 \qquad P(X = 1) = 0.3$$

- Find $E[X^2]$. In order to do this, we need to let $Y = X^2$ and find the pmf for Y.

X	-1	0	1
$Y = X^2$			
$P(X = x)$			

- Next, we need to find the associated probabilities of Y.

Y	0	1
$P(Y = y)$		

- Now we can find $E[Y] = E[X^2]$.

EXPECTED VALUE OF DISCRETE RV'S(CONTINUED)

- Although the previous method will always allow us to compute the expected value of a function of X as long as we know the pmf of X, there is another way of thinking about $E[g(X)]$. Since $g(X) = g(x)$ whenever $X = x$, it seems reasonable that $E[g(X)]$ should just be a weighted average of the values $g(x)$, with $g(x)$ being weighted by $P(X = x)$.

- Proposition: If X is a discrete random variable that takes on the values of x_i where $i \geq 1$, with respective probabilities $f(x_i)$, for any real – valued function g,

$$E[g(X)] = \sum g(x_i)f(x_i)$$

EXAMPLE 6 (CONTINUED)

- Let's make sure this works for the previous example. Recall:

$$P(X = -1) = 0.2 \qquad P(X = 0) = 0.5 \qquad P(X = 1) = 0.3$$

$E[g(X)] = E[X^2] =$

EXPECTED VALUE OF DISCRETE RV'S (CONTINUED)

- Corollary: If a and b are constants, then

$$E[aX + b] = aE[X] + b$$

EXAMPLE 7

- Let X be a R with $f(x)$ as described below:

x	1	2	3	4
$f(x)$	0.40	0.30	0.20	0.10

  - Find $E[x-1]$
  - Find $E[3x+5]$

VARIANCE OF DISCRETE RV'S

- Another characteristic of interest for a discrete random variable is the variance which will provide an indication of variation or spread of the values of X.

- If X is a random variable with mean, μ, then the **variance** of X, denoted by $\mathbf{Var}(\boldsymbol{X})$, is defined by

$$\mathrm{Var}(X) = \sigma_X^2 = E[(X-\mu)^2] = E[X^2] - (E[X])^2$$

VARIANCE: PROOF

- $\mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$
- Proof:

EXAMPLE 8

- Find the variance for the pmf given in Example 3a: $f(x) = \frac{x}{10}$ where $x = 1,2,3,4$

- Find the variance for the pmf given in Example 3c: $f(x) = \frac{1}{30}(x+1)^2$ where $x = 0,1,2,3$

VARIANCE OF DISCRETE RV'S (CONTINUED)

- Corollary: For any constants a and b, then

$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

EXAMPLE 9

- Recall the following from Example 7. Let X be a R with $f(x)$ as described below:

x	1	2	3	4
$f(x)$	0.40	0.30	0.20	0.10

  - Find $\mathrm{Var}(X)$

  - Find $\mathrm{Var}[3x + 5]$

IN-CLASS PROBLEM

- In roulette, a ball rolls around a roulette wheel. 18 are red, 18 are black, and 2 are green. A bet on "red" costs $1 to play. If it lands red, you win $1. If not, you lose your $1.
- What are your expected winnings with this strategy?
- Show your work!

CONTINUOUS RANDOM VARIABLES

CONTINUOUS RANDOM VARIABLE

- We just finished investigating discrete random variables. We are now going to look at continuous random variables.
- A **continuous random variable** is a variable whose set of possible values is uncountable.
  - Not discrete!

- Examples:
  - X = "Time until a train arrives"
  - Y = "Height"
  - W = "Time until failure"

PROBABILITY DISTRIBUTION FUNCTION (PDF)

- A **probability distribution function (pdf)** for a continuous random variable is a function $f(x)$ if the following properties hold:
  - $P(-\infty \leq X \leq \infty) = \int_{-\infty}^{\infty} f(x)dx = 1$
  - All probability statements about X can be answered in terms of f. For instance,

$$P(a \leq X \leq b) = \int_{a}^{b} f(x)dx$$

  - If $a = b$, then $P(a \leq X \leq b) = \int_{a}^{b} f(x)dx = 0$
  - $P(X < a) = P(X \leq a) = F(a) = \int_{-\infty}^{a} f(x)dx$

EXAMPLE 1

- Suppose X is a continuous random variable whose pdf is given by $f(x) = \begin{cases} c(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$
  - Find the value of c.
  - Find $P(X > 1)$

CUMULATIVE DISTRIBUTION FUNCTION (CDF)

- The **cumulative distribution function (cdf)**, $\boldsymbol{F(x)}$, for the continuous random variable X with pdf $f(x)$ is given by

$$F(x) = P(X \le x) = \int_{t=-\infty}^{x} f(t)dt$$

- This implies that $f(x) = \frac{d}{dx}F(x)$

EXAMPLE 2

- Find the cdf for the random variable X whose pdf is given by $f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

EXPECTED VALUE OF CONTINUOUS RV'S

- Recall: For a discrete random variable, the expected value (mean/average) was defined as $E[X] = \sum_x xf(x)$.
- The expected vale for a continuous random variable will be of the same form, except we'll have an integral rather than a summation.

- Thus, the **expected value** for a continuous random variable X can be found by

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

EXAMPLE 3

- Find the expected value for the random variable X whose pdf is given by $f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

EXPECTED VALUE OF CONTINUOUS RV'S

- Proposition: If X is a continuous random variable with pdf $f(x)$, then for any real – valued function $g(x)$,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

- Corollary: If a and b are constants, then $E[aX + b] = aE[X] + b$.

EXAMPLE 4

- Find the $E[X^2]$ for the random variable X whose pdf is given by $f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

VARIANCE OF CONTINUOUS RV'S

- The **variance** can again be found in the following manner:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

EXAMPLE 5

- Find the variance for the random variable X whose pdf is given by $f(x) = \begin{cases} \frac{3}{8}(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

JOINT PROBABILITY DISTRIBUTIONS

JOINT PROBABILITY DISTRIBUTION

- Up until this point, we've been looking at just one random variable at a time. Often, we are interested in the simultaneous outcomes of several random variables

- Examples:
  - For heart patients, we might be interested in heart rate, blood pressure (this could be broken down into systolic and diastolic), and cholesterol (again, could break down to HDL and LDL).
  - For college applicants, we may want to look at high school GPA, class rank, and ACT/SAT score.
  - The white blood cell count for patients at times: 0, 1, and 2 hours after a medicine is administered.

JOINT PROBABILITY DISTRIBUTION OF DISCRETE RV

- The function $\boldsymbol{f(x,y)}$ is a **joint probability distribution** for the discrete random variables X and Y if
  - $f(x,y) \geq 0$ for all (x,y)
  - $\sum_x \sum_y f(x,y) = 1$
  - $P(X = x, Y = y) = f(x,y)$

- These conditions are analogous to what we saw with univariate distributions in the previous two sections.

- As both random variables are discrete, we can also refer to this as a **joint pmf**

EXAMPLE 1

- Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. Let $X =$ the number of red balls drawn and $Y =$ the number of white balls drawn. Find the joint probability mass function of X and Y.
  - Possible values of X are ________________
  - Possible values of Y are ________________

$f(x,y)$		Y			
X					

EXAMPLE 1

$f(x, y)$		Y			
		0	1	2	3
X	0	0.045	0.182	0.136	0.018
	1	0.136	0.273	0.081	0
	2	0.068	0	0	0
	3	0.005	0	0	0

MARGINAL DISTRIBUTION OF DISCRETE RV'S

- Another distribution of interest is the **marginal distribution** which allows us to look at just one of the random variables, instead of jointly.

- The **marginal distribution** can be found by summing the joint distribution over the random variable that you are not interested in. That is,

$$g(X) = \sum_y f(x, y) \qquad h(Y) = \sum_x f(x, y)$$

EXAMPLE 2

- Find the marginal distributions for both X and Y from the joint probability distribution identified in Example 1.

x	0	1	2	3
$g(x)$	0.382	0.491	0.123	0.005

y	0	1	2	3
$\mathrm{h}(y)$	0.255	0.509	0.218	0.018

- Find the probability that at least 2 red balls are drawn.

JOINT PROBABILITY DISTRIBUTION OF CONTINUOUS RV

- The function $\boldsymbol{f(x,y)}$ is a **joint probability distribution** for the continuous random variables X and Y if
  - $f(x,y) \geq 0$ for all (x,y)
  - $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy = 1$
  - $P\big((X,Y) \in A\big) = \int_A \; f(x,y)\; dxdy$ for any region A in the xy plane

- As both random variables are continuous, we can also refer to this as a **joint pdf**

EXAMPLE 3

- Let X be a random variable denoting grade in a math course and Y be a random variable denoting grade in a statistics course with joint pdf of

$$f(x,y) = \begin{cases} x^2 + 2y^2 & x,y \in (0,1) \\ 0 & \text{otherwise} \end{cases}$$

- What is the probability that a student received a grade of B or better (>80%) in the math course and statistics course?

MARGINAL DISTRIBUTION OF CONTINUOUS RV

- The **marginal distribution** can be found by integrating the joint distribution over the random variable that you are not interested in. That is,

$$g(X) = \int_{-\infty}^{\infty} f(x, y)dy \qquad h(Y) = \int_{-\infty}^{\infty} f(x, y)dx$$

EXAMPLE 4

- Obtain the marginal probability of X if

$$f(x,y) = \begin{cases} \frac{2}{3}(x + 2y) & x, y \in (0,1) \\ 0 & \text{otherwise} \end{cases}$$

CONDITIONAL DISTRIBUTIONS

- The idea of a conditional distribution stems from conditional probability. Recall,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- For both discrete and continuous random variables, the **conditional distribution** of Y given that $X = x$ is,

$$f(y|x) = \frac{f(x, y)}{g(x)}$$

- Conditional distributions also satisfy all the properties of a probability distribution. Note that, $g(x) > 0$.

EXAMPLE 5

- Obtain the conditional distribution of Y given X, that is $f(y|x)$ if $f(x,y) = \begin{cases} \frac{2}{3}(x+2y) & x, y \in (0,1) \\ 0 & \text{otherwise} \end{cases}$

INDEPENDENCE

- Two random variables, X and Y are statistically **independent** if an only if

$$f(x,y) = g(x)h(y) \quad \forall (x,y)$$

over which $f(x,y)$ is defined.

- This definition holds for both the discrete as well as the continuous random variables.

EXAMPLE 6

- Are the following random variables, X and Y independent?

$f(x,y)$		Y		$g(x)$
		0	**1**	
X	**0**	0.04	0.16	
	1	0.16	0.64	
$h(y)$				

CHOOSE NOTATION

- Combinations for n choose r objects.

$$\binom{n}{r} = \frac{n!}{(n-r)!\,r!}$$

- Used to determine the number of outcomes/elements in a sample space or event when we are selecting objects where order does not matter and no replacement is allowed.
- Recall, we can then use the number of ways to determine probability, $P(A) = \frac{|A|}{|S|}$.

DOUBLE INTEGRATION & BOUNDARIES

- Consider the joint distribution, $f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
  - Verify the distribution is a valid pdf.

DOUBLE INTEGRATION & BOUNDARIES

- Consider the joint distribution, $f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
  - Find the marginal distribution of Y. (i.e. $h(y)$)

DOUBLE INTEGRATION & BOUNDARIES

- Consider the joint distribution, $f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
  - Find the conditional distribution, $f(x|y = 1)$.

COVARIANCE & CORRELATION

EXPECTED VALUE AND VARIANCE REVIEW

- When dealing with one random variable,
  - We found the expected value by,

$$E[X] = \sum_x x f(x) \qquad \text{and } E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

  - We found the variance (spread) by,

$$\text{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$$

COVARIANCE

- The **covariance** between two random variables X and Y is a measurement of the nature of the linear association between the two. It is given by,

$$\text{Cov}(X, Y) = \sigma_{xy} = E[(X - E[X])(Y - E[Y])]$$

- This can also be expressed as

$$\sigma_{xy} = E[XY] - E[X]E[Y]$$

- Important note: If two variables are independent, then their covariance is 0. This does not always work both ways, that is it does not mean that if the covariance is zero then the variables must be independent.

EXAMPLE 7

- Find the covariance of X and Y, given the joint pdf, $f(x, y) = \frac{2}{5}(2x + 3y)$ for $x, y \in (0,1)$

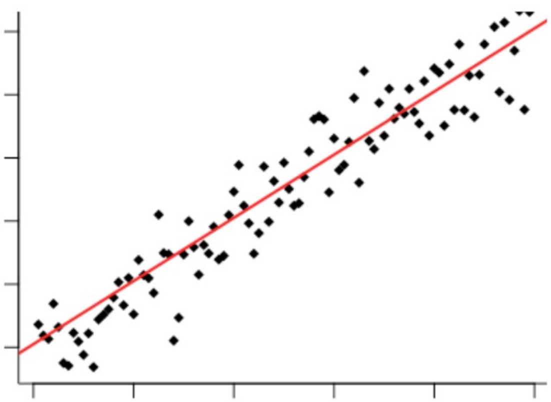
CORRELATION

- **Correlation** describes both the strength and nature (direction) of linear association between two RV's. It is related to the covariance as follows:

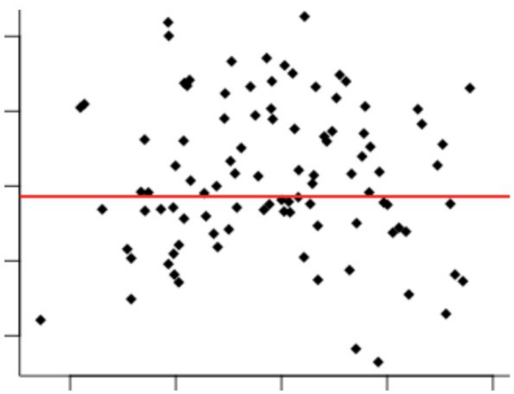
$$\mathrm{Corr}(X, Y) = \rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

- Recognize that $-1 \leq \rho_{XY} \leq 1$

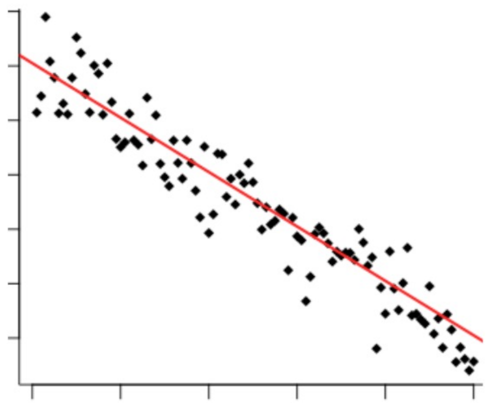
CORRELATION



Positive Correlation ($\rho_{XY} < 0$)



No Correlation ($\rho_{XY} = 0$)



Negative Correlation ($\rho_{XY} > 0$)

CORRELATION

- Correlation has several nice properties:
  - ρ_{XY} is unit-less.
  - $-1 \leq \rho_{XY} \leq 1$
  - The sign of ρ_{XY} determines the direction of the relationship between X and Y.
  - The absolute value of ρ_{XY} determines the strength of the relationship, with larger values indicating less variation (a stronger relationship).
  - If X and Y are independent, then $\rho_{XY} = 0$.

EXAMPLE 8

- Find the correlation between X and Y, given the joint pdf,

$$f(x, y) = \frac{2}{5}(2x + 3y) \text{ for } 0 < x < 1, 0 < y < 1$$

- Comment on the strength of the linear relationship between X and Y.

EXAMPLE 9

$f(x,y)$		Y			$g(x)$
		0	1	2	
X	0	0.1	0.1	0.2	
	1	0.2	0.1	0.1	
	2	0.1	0	0.1	
$h(y)$					

- Find $E[X], E[Y], \mathrm{Var}(X), \mathrm{Var}(Y), \mathrm{Cov}(X,Y), \mathrm{Corr}(X,Y)$

LINEAR COMBINATIONS OF RANDOM VARIABLES

- Let X and Y be random variables
  - $E[a] = a$, where a is a constant
  - $E[aX] = aE[X]$
  - $E[X + Y] = E[X] + E[Y]$
  - $E[aX + bY] = aE[X] + bE[Y]$

  - $\text{Var}(a) = 0$, where a is a constant
  - $\text{Var}(aX) = a^2\text{Var}(X)$
  - $\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$

EXAMPLE 10

- Simplify the expressions below and then evaluate them using
  - $E[X] = 0.8, E[Y] = 1, \text{Var}(X) = 0.56, \text{Var}(Y) = 0.8,$ and $\text{Cov}(X, Y) = -0.1.$

  - $E[3X - 4Y] =$
  - $\text{Var}(3X - 4Y) =$
  - $E[-10X + 2] =$
  - $\text{Var}(-10X + 2) =$
  - $E[1000] =$
  - $\text{Var}(1000) =$